If P is a probability measure on with increasing couplings (see Definition), there exists...

Boubel, Charles, Juillet, Nicolas

doi:10.5802/jep.177; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the frozen source sources/markov-quantile-jep-2022-177/source0.tex; no statement or proof step was rewritten, reordered or removed. References inside the span are handed to the assembler as navigation stubs; load-bearing status is recorded per stub.

Editorial scope. Single lem card cut from doi:10.5802/jep.177 (The Markov-quantile process attached to a family of marginals); statement and proof are the author own text at source lines 460-461 and 1301-1321.

Source reference. Original theorem 1.15: sources/markov-quantile-jep-2022-177/source0.tex, source lines 405–405. navigation stub only: the target lives outside the counted span in the same article; whether it is load-bearing is judged per occurrence

Source reference. Original theorem 1.15: sources/markov-quantile-jep-2022-177/source0.tex, source lines 415–415. navigation stub only: the target lives outside the counted span in the same article; whether it is load-bearing is judged per occurrence

Source reference. Original theorem 3.9: sources/markov-quantile-jep-2022-177/source0.tex, source lines 1034–1034. navigation stub only: the target lives outside the counted span in the same article; whether it is load-bearing is judged per occurrence

Lemma 1.19. *If P is a probability measure on $\mathbb{R}^{\mathbb{R}}$ with increasing couplings (see Definition 1.15*(1.15)*), there exists an increasing process $X = (X_t)_{t\in\mathbb{R}} : \Omega \to \mathbb{R}$, i.e.the functions $t \mapsto X_t(\omega)$ are increasing, such that $P = \mathrm{Law}(X)$.*

Proof of Lemma 1.19. Let $(\mu_t)_{t\in\mathbb{R}}$ be an increasing family of probability measures for $\preceq_{\mathrm{sto}}$ and P be a Markov measure in $\mathrm{Marg}((\mu_t)_{t\in\mathbb{R}})$ such that for every $S = \{s_1, \dots, s_d\}$, the measure $(\mathrm{proj}^S)_{\#}P$ is concentrated on

$$\{(x_1, \dots, x_d) \in \mathbb{R}^d : x_1 \leq \dots \leq x_d\}.$$

Set $\mu_{t^-} := \mathrm{stosup}_{s<t} \mu_s$ and $\mu_{t^+} := \mathrm{stoinf}_{s>t} \mu_s$, that are also the left and right limits of $(\mu_t)_t$ for the weak topology, see Remark 3.9. Since $(\mu_t)_{t\in\mathbb{R}}$ is increasing for $\preceq_{\mathrm{sto}}$, we have $\mu_{t^-} \preceq_{\mathrm{sto}} \mu_t \preceq_{\mathrm{sto}} \mu_{t^+}$ and $t \in \mathbb{R}$ is a discontinuity time of $(\mu_t)_{t\in\mathbb{R}}$ for the weak topology if and only if $\mu_{t^-} \neq \mu_{t^+}$. Such points are

at most countably many. Indeed, the regions in $\mathbb{R}^2$ between the graphs of $F[\mu_{t^-}]$ and $F[\mu_{t^+}]$, for the discontinuity times t, are disjoint and of positive Lebesgue measure, hence are at most countably many, by σ-additivity of the measure.

Let C be a countable dense subset of $\mathbb{R}$ containing the discontinuity points. Introduce:

$$N = \{x \in \mathbb{R}^{\mathbb{R}} : \exists\, (s,t) \in C^2, s < t \text{ and } x(s) > x(t)\}.$$

Being a countable union of P-null sets, N is P-null. Now take $(X_t)_{t\in\mathbb{R}}$ with law P, e.g., take the canonical process $\Omega := \mathbb{R}^{\mathbb{R}}$ and $X_t = \mathrm{proj}^t : x \in \mathbb{R}^{\mathbb{R}} \mapsto x(t)$. We define $(\tilde{X}_t)_t$ as null functions on N, and as follows on $\Omega \setminus N$:

- for $t \in C$, $\tilde{X}_t(\omega) = X_t(\omega)$,
- for $t \notin C$, $\tilde{X}_t(\omega) = \lim_{s<t,\, s\in C} X_s(\omega)$.

Hence for every $\omega \in \Omega$, the curve $t \in C \mapsto X_t(\omega)$ is increasing and, even better, $t \in \mathbb{R} \mapsto X_t(\omega)$ is increasing. We are left to prove that $X_t = \tilde{X}_t$ almost surely. This is clear for $t \in C$. For each $t \in \mathbb{R} \setminus C$, $\{\omega \in \Omega : \exists\, s \in C, s < t \text{ and } X_s(\omega) > X_t(\omega)\}$ is a union of null sets, hence is null, so almost surely, $X_t \geqslant \sup_{s<t,\, s\in C} X_s = \tilde{X}_t$. Besides $X_s \to_{s<t,\, s\in C, s\to t} \tilde{X}_t$ almost surely and thus in law, so that $\mathrm{Law}(\tilde{X}_t) = \mu_{t^-}$. Moreover, t is a continuity point of $(\mu_t)_t$, so that $\mathrm{Law}(\tilde{X}_t) = \mu_t = \mathrm{Law}(X_t)$. Thus, $X_t = \tilde{X}_t$ almost surely. □